

DAY 3

PART 1: INTERVIEW WARM-UP

1. What is a Validation Rule?

A Validation Rule is a Salesforce feature used to check whether the data entered in a record follows the required business conditions. If the condition evaluates to TRUE, Salesforce stops the record from being saved and shows an error message.

Example: Stop a student from applying if their CGPA is below the required CGPA.

2. What is a Flow?

A Flow is a declarative automation tool in Salesforce that helps automate business processes without writing Apex code. It can create or update records, send emails, and perform different actions based on conditions.

Example: Automatically fill the Application Date when a new application is submitted.

3. What is an Apex Trigger?

An Apex Trigger is Apex code that runs automatically when database operations such as insert, update, delete, or undelete occur. It is mainly used when the required business logic is difficult or complex to handle using Flow.

Example: Prevent duplicate applications or update related records automatically.

4. When would you choose a Flow instead of a Trigger?

I would choose a Flow when the requirement can be handled using simple declarative automation without complex coding. Flows are easier to create, understand, and maintain.

Examples:

- Automatically update fields
- Send email notifications
- Create or update related records

5. Can a Validation Rule update another field? Why or why not?

No. A Validation Rule cannot update another field because it is only used to check whether the entered data is valid. It either allows the record to be saved or displays an error message. For automatic field updates, we can use a Flow or Apex Trigger.

6. Which executes first: Validation Rule, Flow, or Trigger?

The simplified execution order for creating or updating a record is:

1. Before-Save Record-Triggered Flow
2. Before Trigger
3. Validation Rule
4. Record is saved
5. After Trigger
6. After-Save Record-Triggered Flow

7. What is a Record-Triggered Flow?

A Record-Triggered Flow is a type of Flow that starts automatically when a record is created, updated, or deleted. It is useful for automating record-related processes without writing Apex code.

Example: When a new Application record is created, automatically set the Application Date and send an email notification. give same questions with another answers

Part 2 – Business Scenario

Scenario:

You are developing a Placement Management System. For each requirement, select the appropriate Salesforce automation tool—Validation Rule, Flow, or Apex Trigger—and explain the reason.

1. Send an email to the Placement Officer when a student applies.

Solution: Flow

Reason: A Record-Triggered Flow can send an email notification automatically whenever a new Application record is created. Since this is a simple automation process, Apex code is not required.

2. Automatically fill the Application Date when a student submits an application.

Solution: Flow

Reason: A Before-Save Record-Triggered Flow can set the Application Date to the current date when the application is submitted. This is an easy and efficient way to update the field automatically.

3. Prevent duplicate applications for the same student and job.

Solution: Apex Trigger

Reason: An Apex Trigger can verify whether an Application already exists for the same student and job before creating a new record. If a duplicate is found, it can stop the record from being saved.

4. Reject applications if the student's CGPA is below the minimum CGPA required for the job.

Solution: Validation Rule

Reason: A Validation Rule can compare the student's CGPA with the minimum CGPA required for the job. If the CGPA is too low, Salesforce prevents the application from being saved and displays an error message.

5. Automatically create an Offer Letter when the Application Status changes to Selected.

Solution: Flow

Reason: A Record-Triggered Flow can identify when the Application Status changes to Selected and automatically create the related Offer Letter record. This can be done easily without writing Apex code.

Summary

In the Placement Management System, different Salesforce automation tools are used based on the requirement. Flow is used for simple automation such as assigning a Placement Officer and sending confirmation emails. Validation Rules are used to prevent incorrect data, such as applications without resumes or changes after the status becomes Placed. Apex Triggers are used for more complex logic, such as updating related records automatically.

Part 3 – Design Challenge

Question: Choose the appropriate Validation Rule, Flow, or Apex Trigger for each requirement and explain the reason.

Requirement	Solution	Reason
Automatically set the Application Date when a new application is created.	Flow	A Before-Save Record-Triggered Flow can automatically set the Application Date before the record is saved.
Prevent a student from applying if their CGPA is below the required minimum.	Validation Rule	A Validation Rule verifies the student's CGPA and prevents the record from being saved if the minimum requirement is not satisfied.
Prevent duplicate applications for the same student and job.	Apex Trigger	An Apex Trigger can check existing records and prevent duplicate applications by using <code>addError()</code> .
Send an email to the Placement Officer after a student submits an application.	Flow	An After-Save Record-Triggered Flow can automatically send an email notification after the Application record is created.
Automatically create an Offer Letter when the application status changes to Selected.	Flow	A Record-Triggered Flow can identify when the status changes to Selected and automatically create the related Offer Letter record.
Update multiple related records with complex business logic.	Apex Trigger	An Apex Trigger is suitable for complex logic involving multiple related records because it provides more control and supports bulk processing.

Part 4 – Hands-on Task Explanation

Objective

Create a **Record-Triggered Flow** for the Placement Management System to:

- Automatically update the Application Status.
- Notify the student about the application.
- Confirm that the flow works successfully.

1. Auto-Update Application Status

Type: Before-Save Record-Triggered Flow

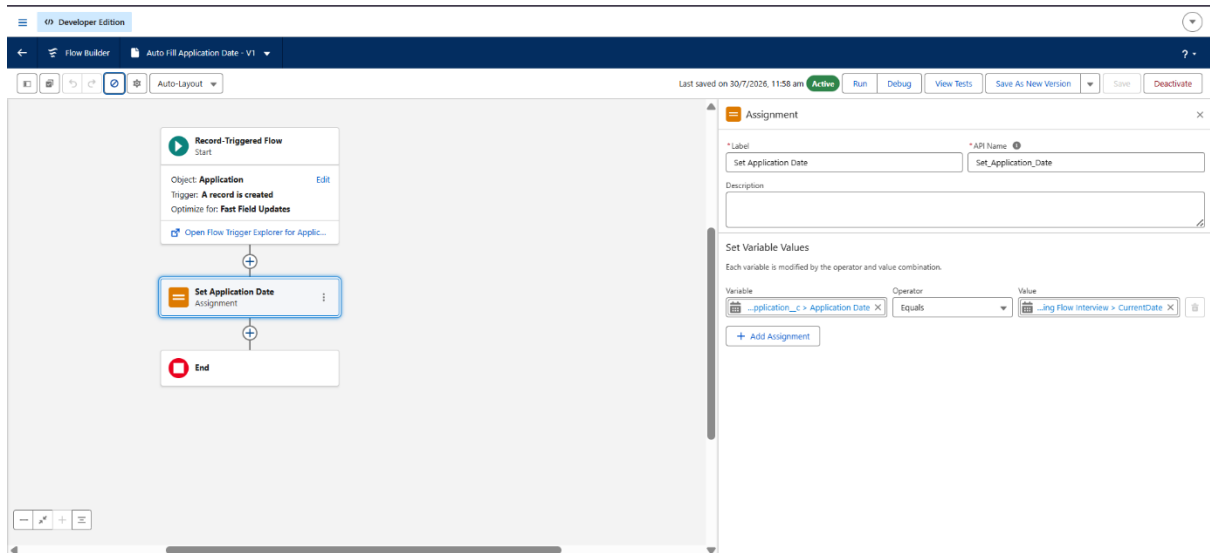
Object: Application

Trigger: When a record is created

Assignment: `Application__Status__c = "Submitted"`

Explanation:

Whenever a student creates a new Application record, the Flow automatically sets the Application Status to **Submitted**. This ensures that all newly submitted applications have the correct initial status without manual updating.



Explanation: Whenever a student creates a new application, the flow automatically fills the Application Date with the current date before the record is saved.

The screenshot shows the Salesforce application record page for the application ID REGD0000000003. The page displays the application details, including the application name, student name, job title, application date, status, and owner. The application date is 30/07/2026 and the status is Applied. The page also shows the 'Related' and 'Details' tabs, and the 'Owner' field is set to Jaya Sri Padilam.

Field	Value
Application Name	REGD0000000003
Student	Jaya
Job	Salesforce Developer
Application Date	30/07/2026
Status	Applied
Created By	Jaya Sri Padilam, 30/07/2026, 12:01 pm
Last Modified By	Jaya Sri Padilam, 30/07/2026, 12:01 pm

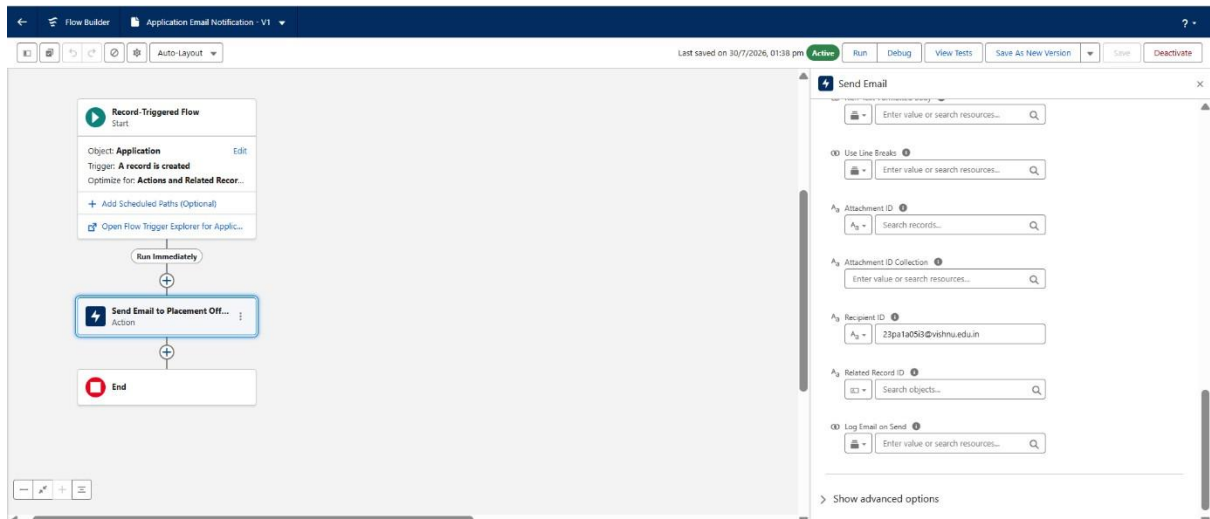
2. Email Notification

Type: After-Save Record-Triggered Flow

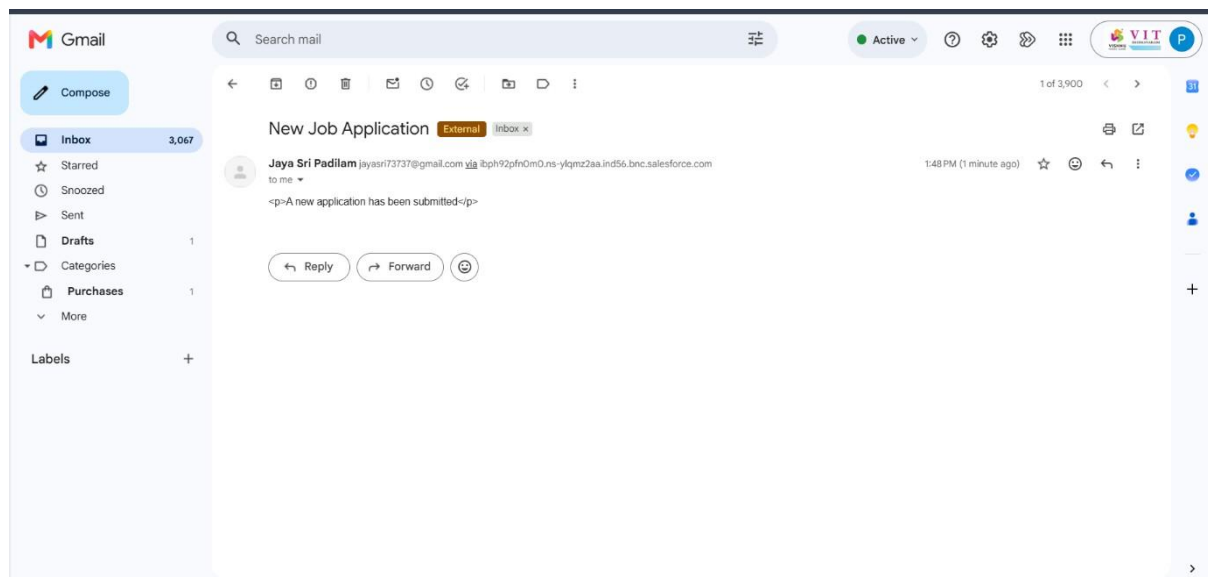
Object: Application

Trigger: When a record is created

Action: Send Email



Explanation: After a student submits an application, the flow automatically sends an email notification to the Placement Officer.



Result: Both flows were created to automate the application process and reduce manual work.

Part 5 – Hands-on Task: Validation Rules

Objective :

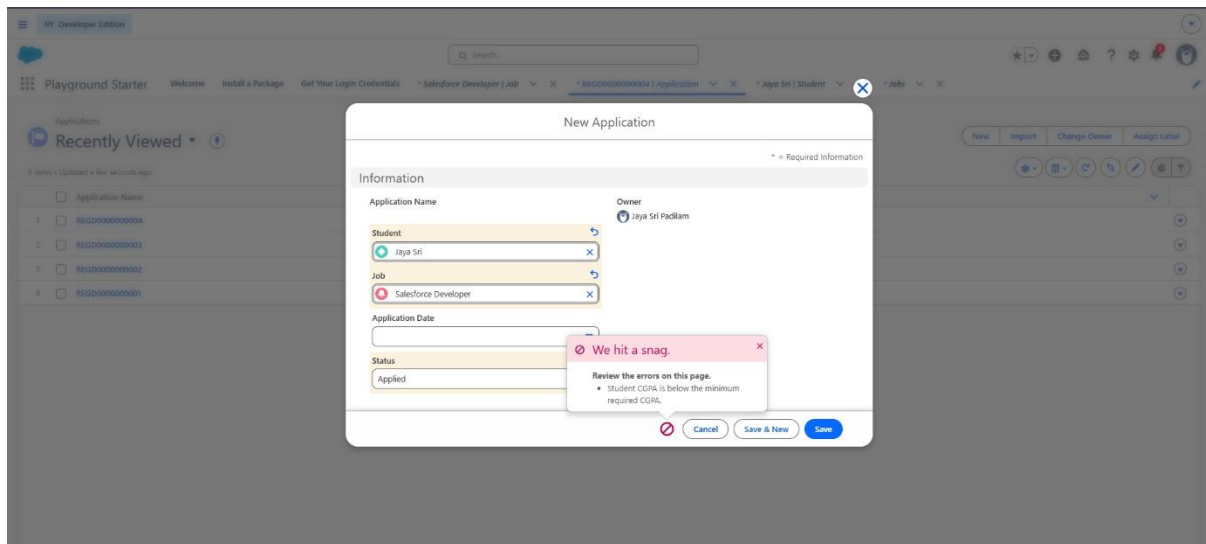
Create Validation Rules on the Application object to prevent invalid application records from being saved and maintain accurate data.

Validation Rule 1 – CGPA Check

Rule Name: CGPA_Check

Purpose: Prevent a student from applying when their CGPA is lower than the minimum CGPA required for the selected job.

Formula: $\text{Student_r.CGPA_c} < \text{Job_r.Minimum_CGPA_c}$



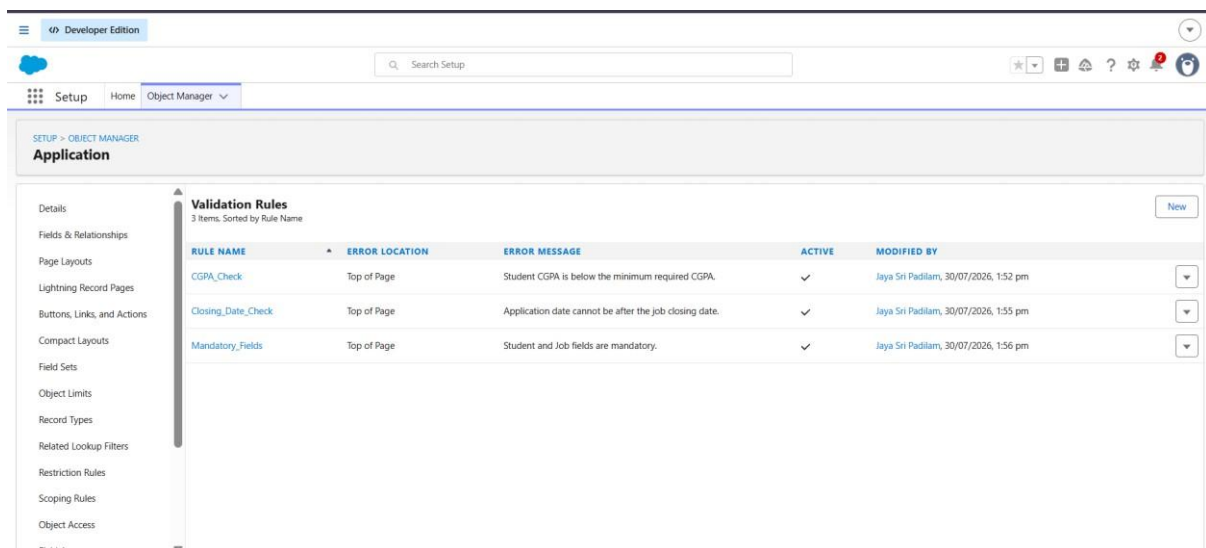
Result: The application cannot be saved if the student's CGPA does not meet the job requirement.

Validation Rule 2 – Closing Date Check

Rule Name: Closing_Date_Check

Purpose: Prevent students from submitting applications after the job's closing date.

Formula: Application_Date__c > Job__r.Closing_Date__c



Result: Applications submitted after the closing date are not accepted.

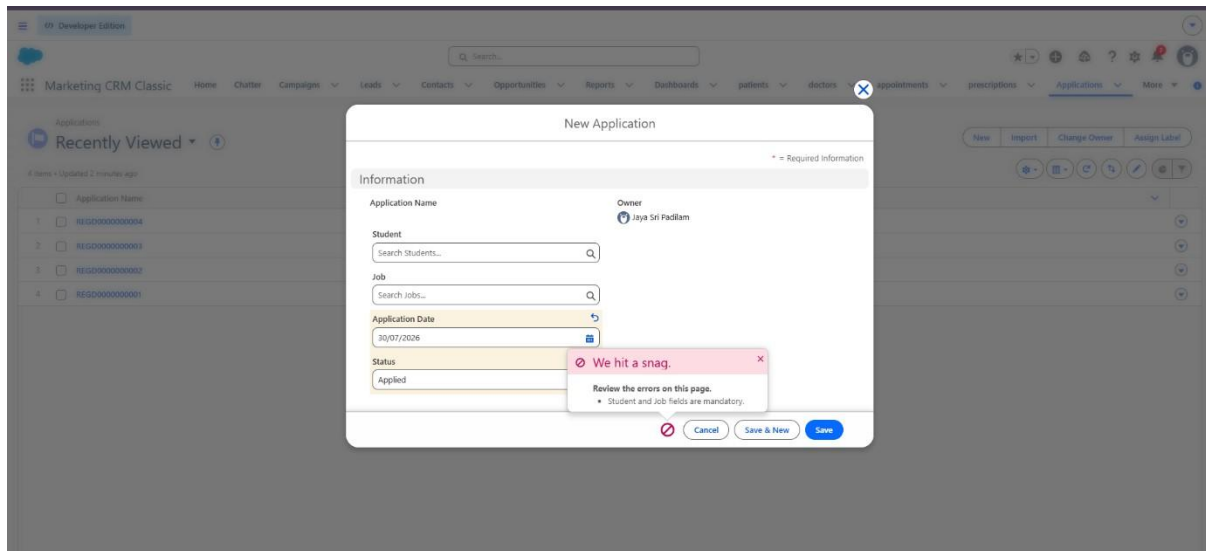
Validation Rule 3 – Mandatory Fields

Rule Name: Mandatory_Fields

Purpose: Make sure that both Student and Job fields are selected before an application is saved.

Formula:

OR(
 ISBLANK(Student__c),
 ISBLANK(Job__c)
)



Result: The application cannot be saved when the Student or Job field is empty.

Part 7 – Mini Project Enhancement

Objective

Enhance the Placement Management System by using Salesforce Flows and Validation Rules to automate placement activities, minimize manual updates, and ensure that application records contain valid information.

1. Automatic Application Status

Implementation:

- Created a Before-Save Record-Triggered Flow on the Application object.
- Configured the flow to execute whenever a new application is created.
- Set the Application Status automatically to Submitted.

Result: Every new application is automatically assigned the Submitted status, which makes application tracking easier.

2. Student Confirmation Notification

Implementation:

- Created an After-Save Record-Triggered Flow on the Application object.
- Configured the flow to send a confirmation email after an application is submitted.
- Used the student's email address as the recipient.

Result: Students receive an automatic confirmation after successfully submitting their application.

3. Validation Rules

The following Validation Rules were configured:

- Email Validation: Ensures that a valid student email address is provided.
- Application Status Validation: Prevents users from entering an invalid application status.
- Required Details Validation: Ensures that important application details are entered before saving.

Result: The Validation Rules prevent incomplete or incorrect records from being saved and help maintain accurate placement data.

4. Automatic Placement Record Creation

Implementation:

- Created an After-Save Record-Triggered Flow on the Application object.
- Configured the flow to run when the Application Status becomes Placed.
- Added a Create Records element to automatically create a related Placement record.

Result: When a student's application status changes to Placed, Salesforce automatically creates the corresponding Placement record, reducing manual work and improving the placement tracking process.

Part 8 – Debugging Challenge**Scenario**

During testing of the Placement Management System, the email notification automation was not working correctly. The purpose was to identify the issue, analyze possible causes, and use Salesforce debugging tools to fix the problem.

Challenge 1 – Email Notification Not Sent**Problem:**

The Placement Officer did not receive an email notification when a student submitted a new Application.

Possible Causes:

- The After-Save Record-Triggered Flow may not be activated.
- The Placement Officer's email address may be missing or incorrect.
- The Flow entry conditions may not be satisfied.
- The Send Email action may not be configured correctly.
- Salesforce email deliverability settings may restrict outgoing emails.

Debugging Steps:

1. Open the Flow in Flow Builder and check the trigger conditions.
2. Verify that the Flow runs when a new Application record is created.
3. Check the Placement Officer's email address.
4. Verify the configuration of the Send Email action.
5. Use the Debug option in Flow Builder to test the automation.
6. Check Salesforce email deliverability settings.
7. Correct the identified issue and activate the latest Flow version.

Expected Result:

Whenever a student submits a new Application, the Placement Officer should automatically receive an email notification containing the application details.